

Evaluating Media Equality in Generative Audiovisual Composition Using Modality Constraint and Replay-Based Reveal

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Abstract

Coupled audiovisual instruments, in which sound and image are produced from a shared generative process, promise media equality by design. But does causal coupling actually deliver perceptual parity? This article asks how composers experience media equality and intermedial interference when working with such an instrument.

We report a practice-based study of the Cellular Au-Tonnetz, a system in which audio and LED light are parallel projections of the same evolving cellular-automaton state. Nine participants composed short pieces under three modality conditions: visual-only, audio-only, and audiovisual. Each was followed by replay with both modalities revealed. By withholding one medium during composition and reinstating it at replay, the protocol makes the construction of audiovisual experience observable.

Audiovisual access was preferred overall (8 of 9 participants), but the audio-only condition most often produced the largest perceived mismatch (6 of 9), and replay frequently turned apparent mismatch into coherent re-reading. Mismatch typically took the form of cue conflict: temporal persistence, density correspondence, perceptual masking. Productive interference emerged at replay through rebinding. We position this as foundational practice-based work, offering a method other practitioners can adopt, and design priorities for instruments where sound and image share a single generative substrate.

1 Introduction

Electroacoustic audiovisual composition is often discussed through models that implicitly privilege one medium: sound as structural driver with image as visualisation, a stance common in visual music practice (Hyde 2012; Watkins 2018); or image as narrative driver with sound as enhancement, inherited from film sound (Chion 1994; Cook 1998). Even where practitioners aim for parity, perceptual salience and the practical realities of workflow and interface design tend to reintroduce hierarchy, a tension that Harris (2016) and Hill (2013) have articulated explicitly. The Organised Sound call for this thematic issue highlights this as a problem of media equality and raises intermedial interference as a prompt for reconceptualising how media relationships are composed and perceived (Organised Sound 2026). Interference has also been developed as a compositional and analytic focus within electroacoustic audiovisual practice (Chiaromonte 2024).

The article takes a design-and-evaluation approach. We treat intermediality as a manipulable design problem that can be evaluated in practice, focusing on generative instruments in which sound and moving image are produced in real time from shared processes. In such instruments, composing means shaping the dynamics of an underlying generative substrate rather than synchronising independent media layers after the fact.

The coupling pattern we examine is state-space projection: sound and image are parallel projections of a single evolving state, rather than one medium being derived from or aligned to the other. Because the coupling is causal, with every change in the underlying state changing both modalities, it provides a concrete basis for evaluating intermedial interference.

1.1 Intermedial interference as a design and evaluation problem

Intermedial interference is used here, following Chiaramonte (2024), to denote the perceptual and interpretive consequences of interactions between salient features of the two media. It includes constructive effects (reinforcement, increased legibility, and meaningful surprise) and destructive effects (contradiction, masking, overload, and reduced causal confidence). In a state-space projection system, interference is design-sensitive: it emerges from mapping choices, temporal envelopes, constraint regimes, and the performer’s ability to build and test a causal model while composing.

Despite the prevalence of generative systems in electroacoustic and audiovisual practices, evaluation is often limited to descriptive accounts or anecdotal response. This makes it difficult to compare design choices or to articulate how a system supports or undermines parity between media. The field needs evaluation approaches that combine practice-based validity with analysable evidence and that link participant experience to perceivable properties of captured artefacts.

1.2 Research questions and contributions

The study addresses three questions:

RQ1 (intermedial equality): To what extent do participants perceive sound and light as co-equal contributors to the audiovisual experience when composing with the system?

RQ2 (intermedial interference): What forms of constructive and destructive intermedial interference arise in practice, and how do they relate to modality access and parameter shaping?

RQ3 (agency and causality): How do perceptions of authorship, predictability, and steerability change when one modality is unavailable during composition?

The article contributes:

- A state-space projection framing of intermedial composition in which sound and light are treated as parallel projections of a shared generative substrate, enabling analysis without default media hierarchy.
- An evaluation method for practice-based intermedial research, combining modality-access manipulation, replay-based reveal, and traceable artefact capture, that produces analysable evidence within realistic artistic development timelines.

- An interference-focused account of mechanisms (cue conflict and rebinding) and instrument-actionable design priorities that improve media equality in coupled generative instruments.

2 Related work

Intermediality is often used as a general label for work between media, but its analytic value depends on specifying the intermedial relation at stake. Rajewsky distinguishes medial transposition, media combination, and intermedial reference; this paper is concerned with media combination and, specifically, whether sound and time-based light output are experienced as co-constitutive rather than hierarchical (Rajewsky 2005).

Elleström’s modality model provides a useful scaffold for discussing equality without treating media as undifferentiated streams. By distinguishing material, sensorial, spatiotemporal, and semiotic modalities, the model makes it possible to describe where equality succeeds or fails: parity may hold in causal origin while collapsing in sensorial salience or semiotic authority (Elleström 2010).

Audiovisual theory in film and sound studies provides durable concepts for how sound and image are perceived as an integrated whole. Chion’s account of audio-vision emphasises that audiovisual perception is not additive; sound and image reciprocally frame one another’s meaning. This is directly relevant to a modality-withholding design: composing with one modality absent and reinstating it at replay makes the construction of audio-vision observable, including how expectation, surprise, and retrospective binding contribute to perceived unity (Chion 1994).

Beyond Chion, several frameworks address the encounter and composition of electroacoustic audiovisual works directly. Coulter (2010) classifies sound-image pairs along referential and abstract axes and distinguishes concomitant from isomorphic relationships, offering a vocabulary for how shared formal characteristics catalyse audiovisual integration. Garro (2012) places audiovisual association strategies on a continuum from separation through intuitive complementarity, synchresis, and parametric mapping, contrasting Chion’s visually-driven cinema contract with a visual music contract that holds “the primacy of both ear and eye together” as its artistic credo. Hill (2013) extends this work towards audience interpretation, where schemata, lived experience, and contextual framing make audiovisual works legible across diverse encounters. Harris (2016) takes an explicit anti-hierarchy position, rejecting the term “visual music” because it implies musicality takes precedence and arguing instead for sound and image as parts of a single coherent system. From within the visual music tradition itself, Hyde (2012) develops the concepts of visual silence, visual noise, and visual suspension as ocular counterparts to *musique concrète*, and Watkins (2018) treats audio-visual rhythm as a compositional concern.

Within electroacoustic and intermedial discourse, intermedial interference foregrounds constructive and destructive interactions between media features and provides a handle for theorising media equality beyond simple synchrony (Organised Sound 2026; Chiamonte 2024). Remediation and multimedia analysis further motivate treating media relations as constitutive rather than additive, and provide vocabulary for how one medium can frame or re-situate another (Bolter and Grusin 1999; Cook 1998).

For coupled generative instruments, crossmodal correspondences offer a complementary design lens. Spence synthesises evidence that people reliably associate features across modalities (for example, pitch and brightness), making correspondences a resource for designing legible mappings (Spence 2011). In the present system, correspondence helps explain why some participants use colour similarity as a proxy for harmonic planning even without formal theory.

Evaluation traditions in digital musical instruments provide precedents for studying agency negotiation without reducing outcomes to usability. O’Modhrain proposes a framework supporting structured reflection on experience while remaining sensitive to musical context (O’Modhrain 2011). Orio and Wanderley similarly discuss method rationales for evaluating input devices for musical expression that bridge performer experience and system iteration (Orio and Wanderley 2002).

Finally, the specific substrates used here have well-established lineages. Tonnetz representations are canonical in neo-Riemannian and transformational theory, where neighbourhood relations correspond to parsimonious voice-leading operations (Cohn 1997; Douthett and Steinbach 1998). Cellular automata also have a long-standing lineage as musical generative formalisms valued for emergent structure from simple local rules (Miranda 2007). Besada et al. highlight cognitive and HCI challenges in Tonnetz-at-first-sight interaction, where visible patterns in the lattice geometry, such as the triangular groupings that form familiar chord shapes, can bias harmonic expectation; here, such bias is treated not only as a usability risk but as a potential resource for productive intermedial interference in a coupled audio and light system (Besada et al. 2024).

3 Method

3.1 Participants and ethics

Nine participants completed the solo protocol ($N = 9$), recruited via convenience sampling across a range of musical experience and familiarity with generative systems. Participants provided informed consent for audio and video capture and for anonymised excerpts to be used as research stimuli. Background questions captured self-reported musical experience, theory familiarity, familiarity with generative systems, Tonnetz familiarity, and any relevant perceptual notes (for example, colour-vision deficiency and sensitivity to flashing lights).

3.2 System under study

The study evaluates the Cellular Au-Tonnetz (TZ5), previously described as a unified audiovisual MIDI generator that couples Tonnetz pitch-space mapping with cellular automata and a web-configured embedded control architecture (Didiot-Cook 2025).

The system’s core design feature is shared causal origin. A Tonnetz-mapped lattice of pitch classes evolves via asynchronous cellular automata dynamics under local rule thresholds and global population constraints. The same evolving state is rendered as (i) MIDI-driven audio output and (ii) LED-panel activity using a fixed pitch-class-to-colour mapping. In this framing, sound and light are parallel projections of a shared latent score: a single evolving substrate that is perceived through two modalities.

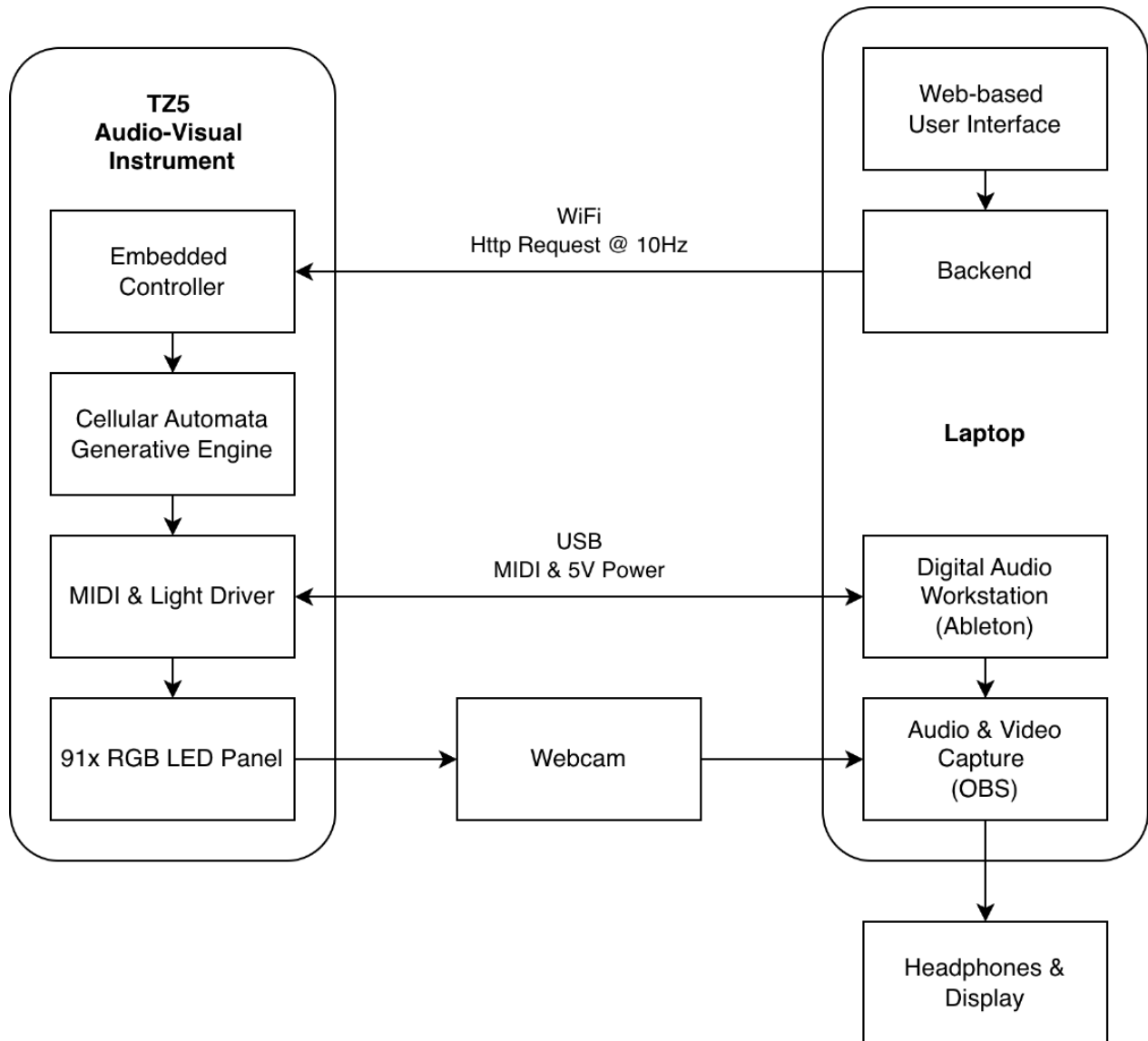


Figure 1: System block diagram, comprising the TZ5 instrument and its constituent parts, as well as the supporting infrastructure and instrumentation required to conduct the study.

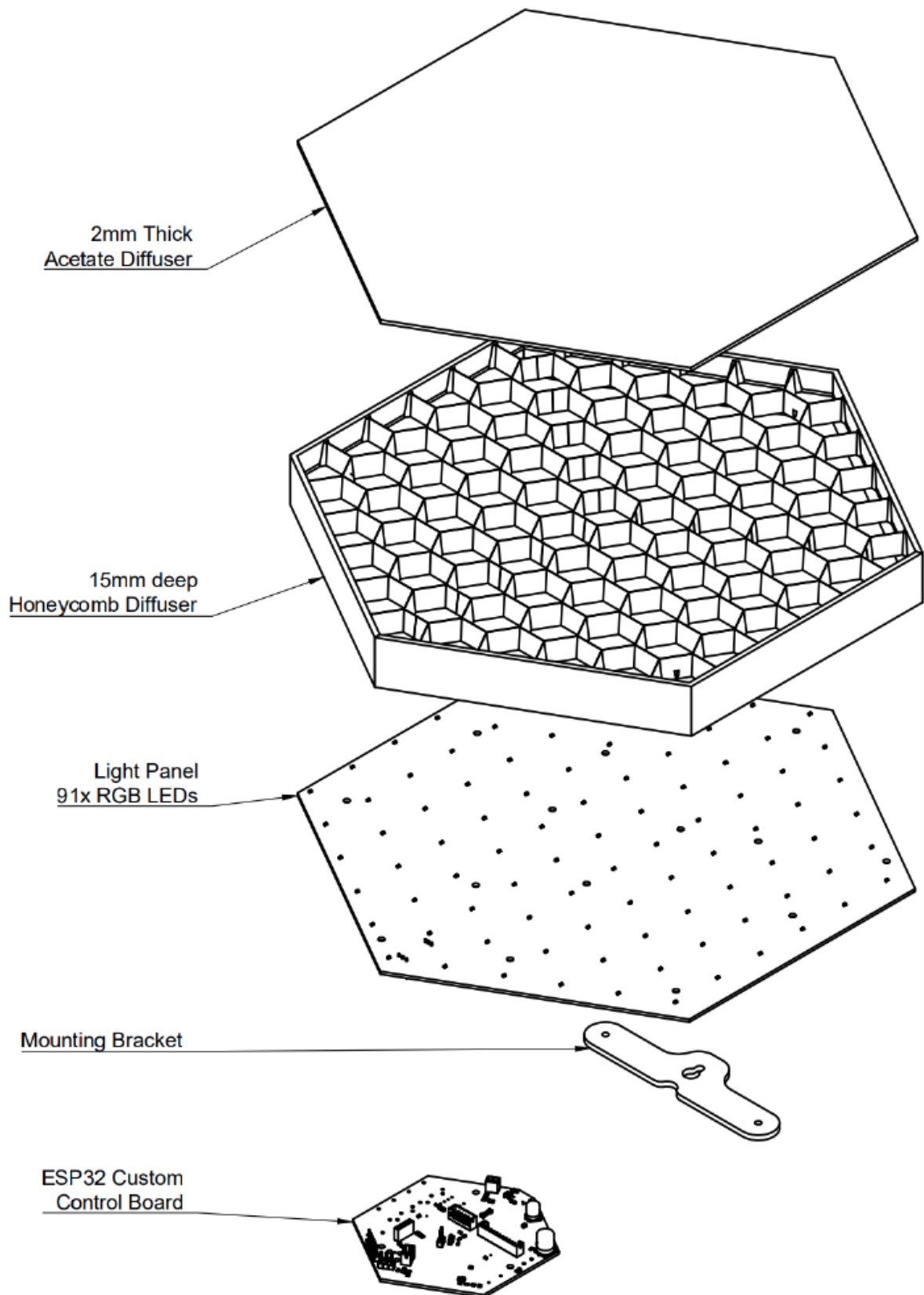


Figure 2: Exploded view of TZ5 audiovisual instrument. Light from the 91x Red, Green & Blue (RGB) LEDs is guided by a hexagonal lattice and diffused by a 2mm sheet of acetate.

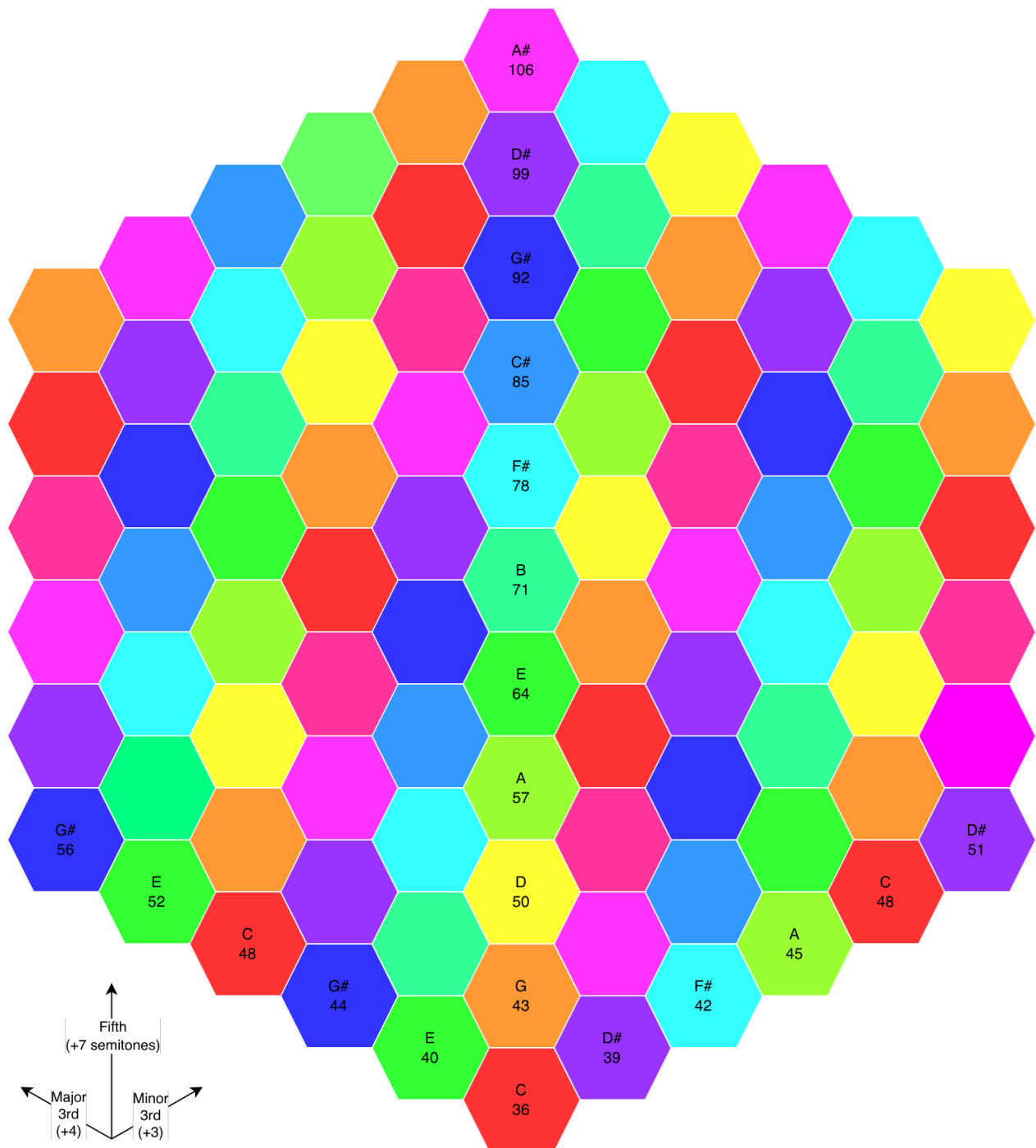


Figure 3: Tonnetz pitch-class map used for hue mapping. MIDI numbers are given under each pitch class, and computed from the root of C;36 (bottom, centre), incrementing by +3 semitones moving right, and +4 moving left.

3.3 Participant-facing control parameters

Participants steered the system exclusively via a browser-based user interface. The parameter set was intentionally compact and targeted density, temporal dynamics, neighbourhood constraints,

recurrence, and harmonic admissibility:

- **Density bounds:** Min Population and Max Population.
- **Temporal dynamics:** Rate (update interval) and Life Length (activation lifetime).
- **Neighbourhood constraints:** Neighbourhood extent (6 local neighbours or 18 extended), Min Neighbours, and Max Neighbours.
- **Recurrence:** Loop On or Off and Loop Length.
- **Harmonic admissibility:** Scale (constrains pitch classes and therefore the visible palette).

3.4 Study design

The solo study used a within-subject design with three modality-access blocks per participant:

A: visual-only composition (panel visible; audio masked)

B: audio-only composition (audio available; panel occluded)

C: audiovisual composition (both modalities available)

In all blocks, both audio and video were recorded and the end-state parameter preset was saved for traceability and repeatability. Block order was counterbalanced across participants.

3.5 Procedure and measures

Each block began from a shared starting preset (S0) and lasted approximately 5 minutes. Participants were instructed to create something interesting, and to record a final excerpt (60 to 120 seconds) with parameter changes permitted during recording. Immediately after each capture, the excerpt was replayed with both modalities enabled (the reveal), followed by a two-part questionnaire:

Part A (pre-reveal): ratings and brief notes about the composition experience under constraint.

Part B (post-reveal): ratings and reflections about the fused audiovisual result after replay.

Likert items used a 7-point scale (1 = strongly disagree, 7 = strongly agree). Part A captured satisfaction (A1), intention clarity (A2), steerability (A3), interface understandability (A4), useful surprise (A5), frustrating unpredictability (A6), and confidence others would find the result interesting (A7). Part B captured same-process judgement (B1), modality balance (B2), coherence (B3), constructive reinforcement (B4), destructive contradiction (B5), overload (B6), expectation match (B7), interpretation change (B8), causal story plausibility (B9), perceived autonomy (B10), reliance on visual cues (B11), and reliance on theory cues (B12). After each block, participants nominated up to three parameters they perceived as most influential.

At session end, participants ranked their three outputs, selected the condition that felt most intermedial, selected the condition with the biggest mismatch, and described one change that would improve media equality or intermedial legibility.

Where participant quotations appear in this article, minor spelling and typing errors have been silently corrected for readability; meaning and structure are unchanged.

4 Results

4.1 Data completeness

All nine participants completed all three blocks (27 of 27 blocks). Saved presets provide a direct link from questionnaire responses to reproducible system states.

4.2 Condition preference, perceived intermediality, and mismatch

End-of-session rankings show a strong preference for audiovisual access: 8 of 9 participants ranked condition C as best overall; 1 of 9 ranked condition B best overall. Participants most frequently selected condition C as most intermedial (7 of 9), with one selecting B and one indicating uncertainty.

Mismatch reports concentrated in the audio-only condition: 6 of 9 participants nominated B as the biggest mismatch, compared with A (2 of 9) and C (1 of 9).

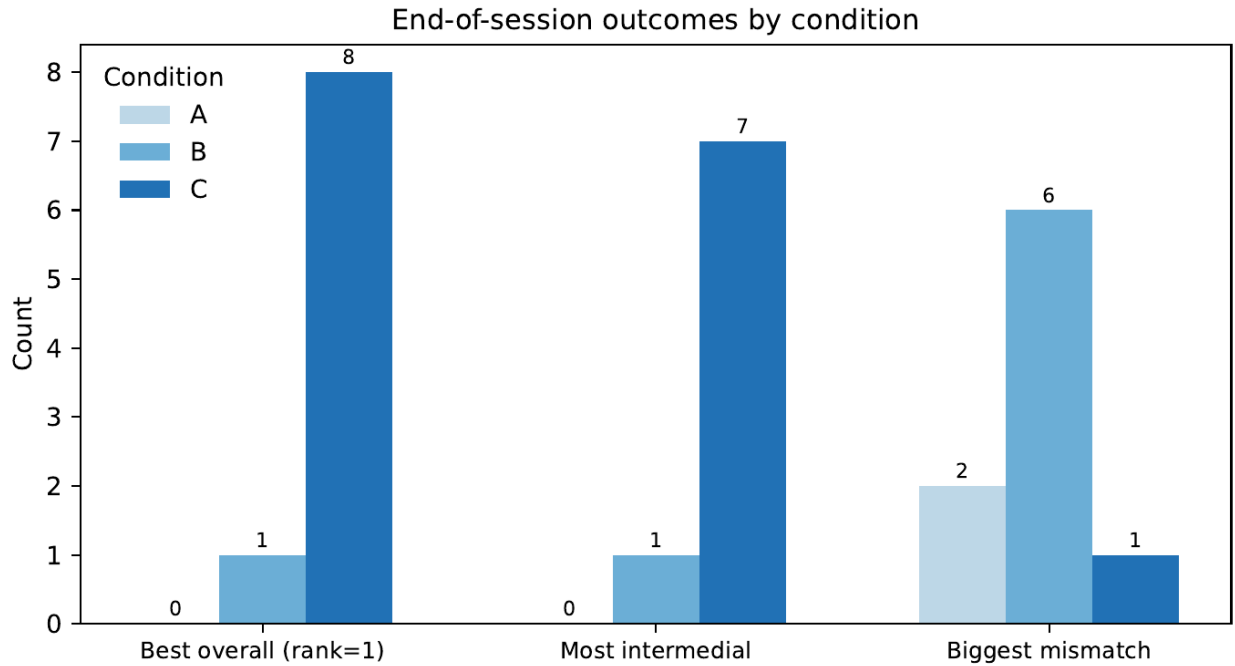


Figure 4: Counts of best overall rank, most intermedial selection, and biggest mismatch by block condition (A=Visual Only, B=Audio Only, C=Audiovisual).

4.3 Strategies under constraint

Participants described compositional approaches that clustered into strategy families: density shaping, arcs or narrative forms, loop-based structuring, scale or harmony selection, and rapid A/B testing of parameters to learn local cause-effect relations.

Visual-only composition often supported a stable action-evaluation loop. Figure 5 shows A2 (intention clarity) and A3 (steerability) clustering towards agreement in condition A relative to condition B, consistent with state visibility supporting intention formation and monitoring. For

example, one participant aimed to fill the grid and shape form over time: “Fill the whole grid with lights. Build up a full grid then switch the rate up and down” (Participant 3, visual-only block, pre-reveal). Another described a similar density-first approach: “Start with lots of colour covering the grid, then reduce intensity and bring it back up to full coverage” (Participant 1, visual-only block, pre-reveal).

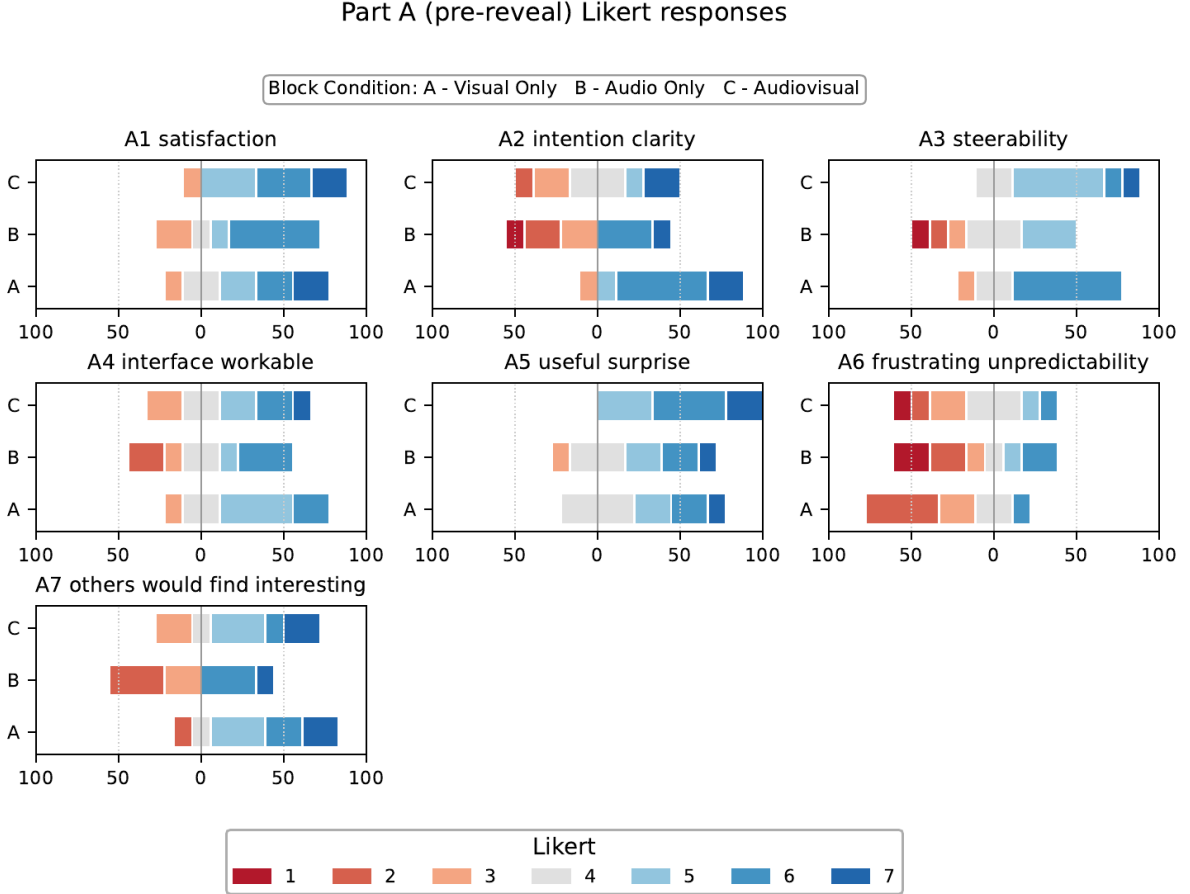


Figure 5: Diverging stacked Likert responses for Part A (pre-reveal) items by block condition. Bars show within-condition percentages centred on neutral (4).

Visual-only access was sometimes described as creatively liberating, especially for trained musicians. Participant 8 reported that removing audio reduced theory-led self-monitoring and enabled a more playful mode of exploration: “didn’t even think about pitch and what scale notes to use on the visual block... satisfaction is interesting as I felt in a playful state so satisfaction higher than if I was in a performance /product state” (Participant 8, visual-only block, post-reveal).

Audio-only composition shifted strategies towards pacing, texture, and repetition as a stabiliser. Figure 5 shows A2, A3, and A4 (interface understandability) more mixed in condition B, consistent with the loss of visible state during action. Participants responded by creating musical containers through recurrence: “Looping was fundamental... starting with basic, empty loop... keeping it

sparse... and then come back to another loop” (Participant 8, audio-only block, pre-reveal). Others described exploratory testing: “More A/B testing to understand the properties” (Participant 3, audio-only block, pre-reveal).

Audiovisual composition commonly integrated lessons learned from constrained runs. Figure 5 shows high satisfaction (A1) and high useful surprise (A5) in condition C, alongside qualitative accounts of combining purposeful shaping with exploratory play. One participant also noted a practical attentional tension: “had to remind myself to keep looking at the visuals... more of an affinity with sound” (Participant 8, audiovisual block, pre-reveal).

4.4 The reveal as a site of rebinding and reinterpretation

A recurring theme was the reveal acting as a moment of retrospective binding. Figure 6 shows that B1 (same-process judgement) and B4 (constructive reinforcement) are strongest after audio-only and audiovisual composition, while B8 (interpretation change) increases most when one modality has been withheld, consistent with replay functioning as calibration.

Part B (post-reveal) Likert responses

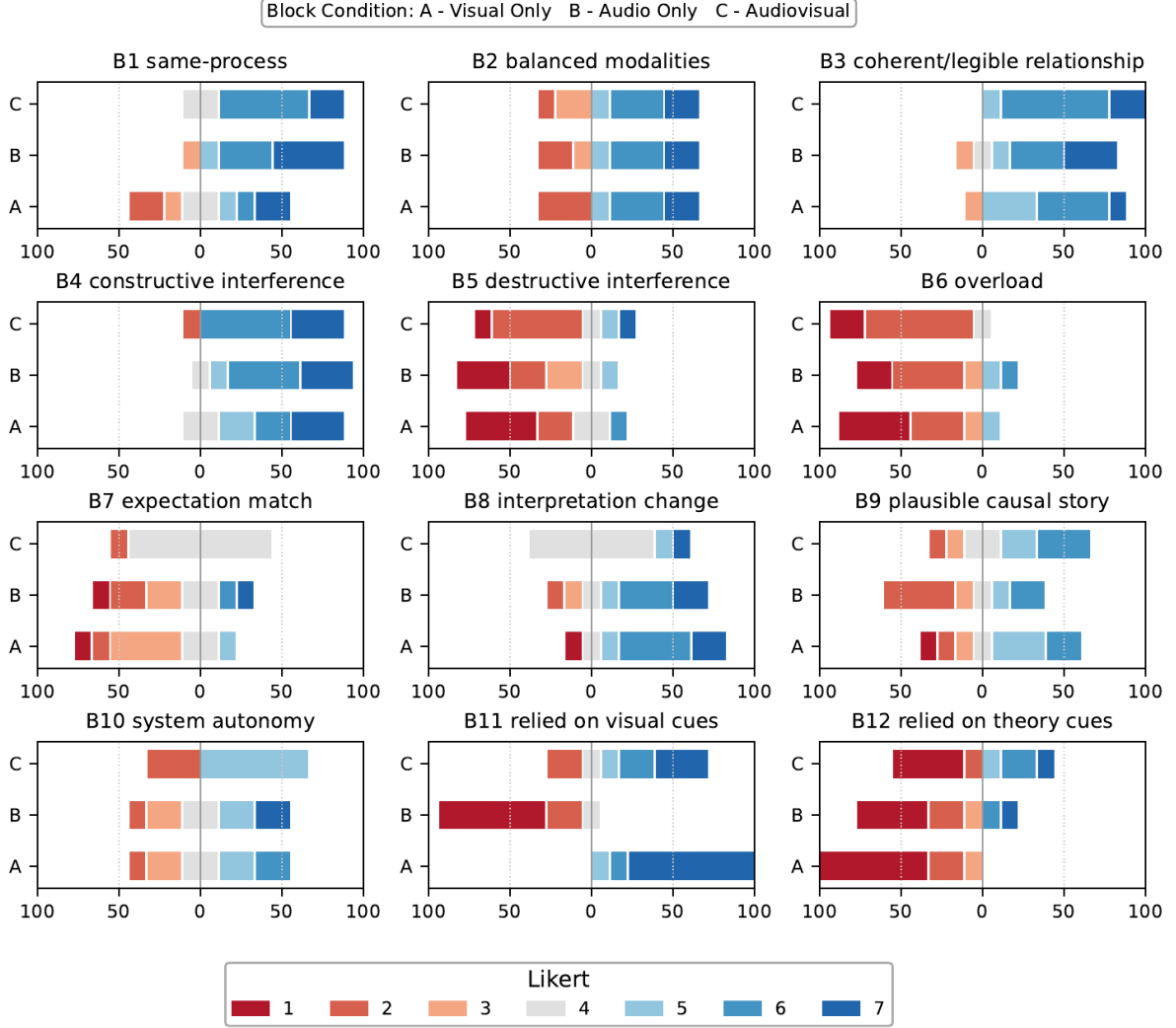


Figure 6: Diverging stacked Likert responses for Part B (post-reveal) items by block condition. Bars show within-condition percentages centred on neutral (4)

Qualitative accounts support this. One participant described audio-only replay as exceeding expectations: “The outcome was more satisfying than I felt the recording and playing process went... on review had more structure than maybe I realised during the process of recording” (Participant 3, audio-only block, post-reveal). Another described a strong positive mismatch on reveal: “I had no expectation of what the visuals should be, but when I saw it, it was so beautiful” (Participant 8, audio-only block, post-reveal).

4.5 Attentional hierarchy and media equality as negotiated practice

Participants repeatedly described media equality as something they had to achieve, rather than something guaranteed by coupling. Several accounts describe sound as perceptually dominant and

therefore hard to ignore: “For me, the audio had to be removed completely before I could prioritise the visual. The audio was more important to my overall experience” (Participant 2, end-of-session reflection). Others described the inverse: “The visual patterns help reveal the relationships between notes” (Participant 3, audio-only block, post-reveal).

These experiences align with cue reliance patterns. Figure 6 indicates that reliance on visual cues (B11) remains substantial even when the visual channel was absent during composition, suggesting that the reveal can re-weight the causal model towards visual structure when it becomes available.

4.6 Intermedial interference mechanisms as cue conflict

When interference was described as disruptive, it was typically framed as cue conflict where one modality implied temporal, structural, or causal relations that the other did not support.

Duration persistence: “Length of light activity was misleading for tone length which was shorter and did not match” (Participant 4, visual-only block, post-reveal). This weakens temporal correspondence and predictive power.

Density correspondence: “A number of sparse visuals sounded as busy as a previously very active visual” (Participant 9, visual-only block, post-reveal). This undermines causal confidence in cross-modal heuristics.

Perceptual masking and auditability: “Different octaves of the same note starting at the same time sounded as one” (Participant 9, visual-only block, post-reveal). When events are not audibly separable, cross-modal checking is weakened.

Participants also reported productive interference when cues reinforced, reflected in Figure 6 by strong endorsement of B4 (constructive reinforcement) and comparatively lower endorsement of B5 (destructive contradiction) in the audiovisual condition.

4.7 Agency and co-creative negotiation

Across conditions, participants framed interaction less as direct control and more as negotiated agency. Some described collaboration with the instrument: “there’s something like a collaboration working with the instrument” (Participant 8, audiovisual block, pre-reveal). Others framed the experience as discovering structure beyond their own intention: “The reviewed videos were more satisfying than I might have imagined... having played a part in constructing the light and sound felt like I had some agency and ability to control a somewhat autonomous system” (Participant 3, end-of-session reflection).

These accounts align with Figure 5 where A5 (useful surprise) remains high across conditions, while A6 (frustrating unpredictability) is not uniformly dominant, indicating that emergence was often valued when it remained steerable.

4.8 Parameter strategy and perceived leverage

Across conditions, participants identified a stable core of influential controls and a modality-dependent shift. Rate was consistently nominated as influential, aligning with qualitative

descriptions of shaping overall dynamism and arc structure. Scale was nominated more frequently when sound was available, consistent with participants treating harmonic admissibility as a primary steering dimension when auditory feedback is present.

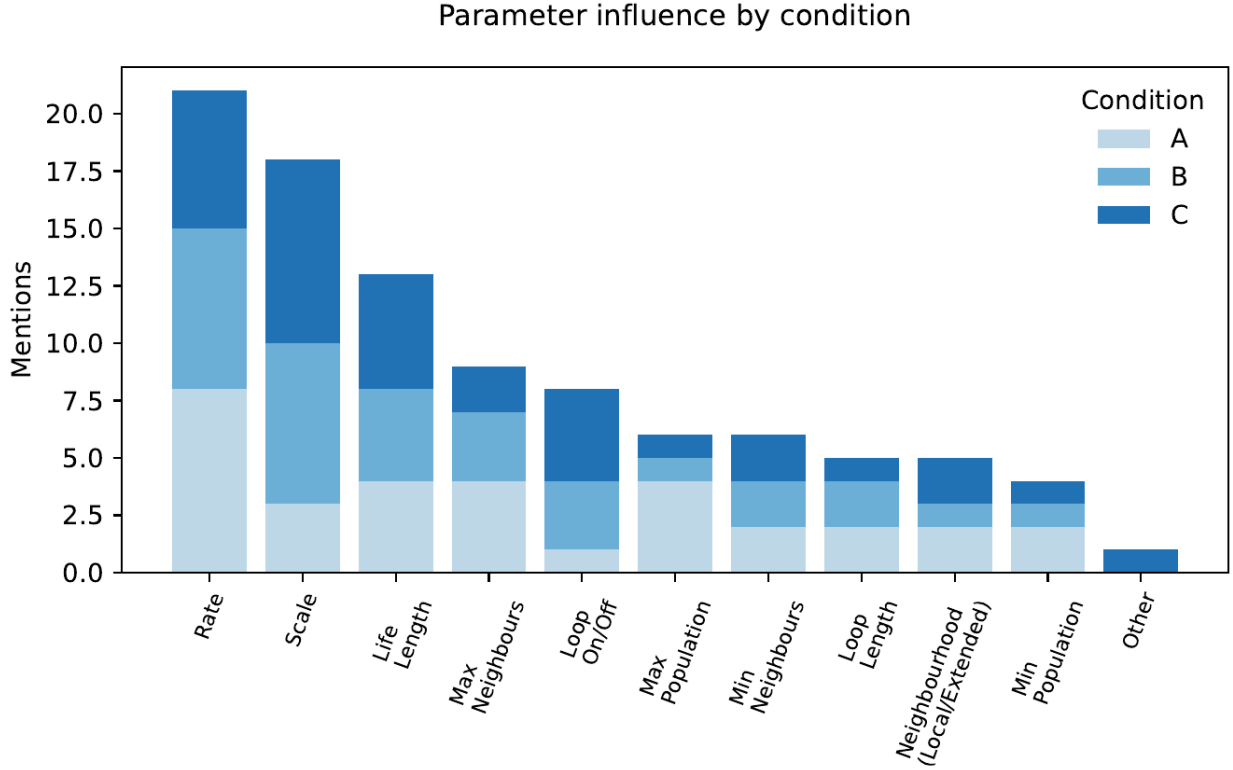


Figure 7: Stacked bar chart of parameter influence nominations by condition.

4.9 Participant-led design requirements

Participant change requests converge on cue reliability and constraint transparency:

- **Align temporal cues:** “Incorporate ADSR envelopes into the audio and visuals” (Participant 1, end-of-session reflection).
- **Improve mapping legibility:** “Seeing the colours of the notes, regardless of them being played may have helped build more harmonious sounds” (Participant 3, end-of-session reflection).
- **Reduce overload and support attention management:** “just having high activity/dynamics of light and sound would be overwhelming and not satisfying” (Participant 4, end-of-session reflection).
- **Improve interface ergonomics:** “Put the interface on one page without scrolling” (Participant 9, end-of-session reflection).

These requirements map directly onto the interference mechanisms above: where cue structure was reliable, participants reported reinforcement; where cues conflicted or masked one another,

participants reported mismatch and reduced steerability.

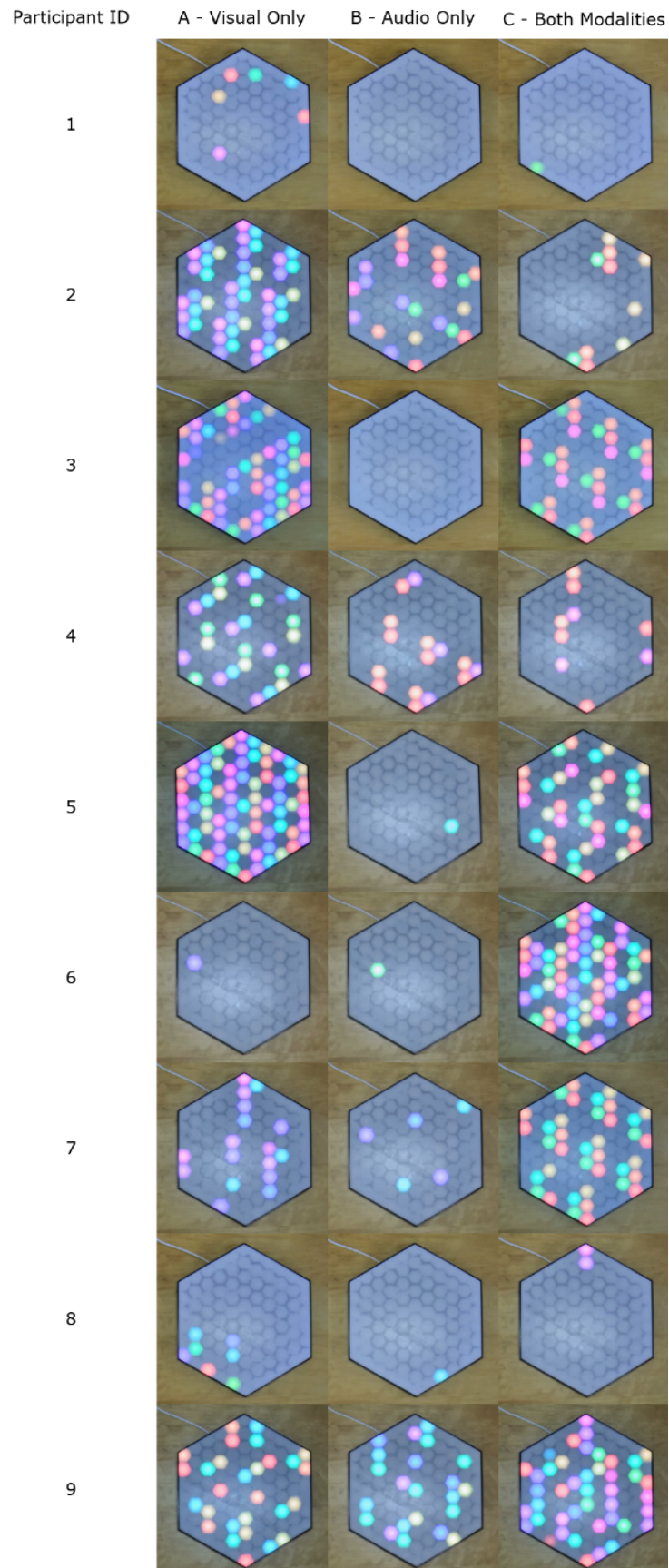


Figure 8: Montage of video-stills taken from participant outputs across conditions (A/B/C).

5 Discussion

5.1 Media equality is not guaranteed by coupling

The Cellular Au-Tonnetz represents sound and light as parallel projections of a shared evolving state. However, participant reports show that causal parity does not automatically yield perceptual or interpretive parity. Equality emerged as negotiated practice shaped by attentional dominance, cue reliability, and participants’ evolving causal models. The strong preference for audiovisual access indicates that simultaneous access most often supports the full intention-monitoring-interpretation loop: participants can form intentions with reference to both media, test hypotheses during interaction, and refine causal inference in real time.

Modality constraint, applied within a coupled system, can be beneficial rather than only degradative. One trained musician described visual-only composition as creatively liberating because it suspended real-time harmonic self-monitoring and invited a more exploratory stance. Temporarily relinquishing one modality can reduce evaluative pressure and open alternative compositional strategies, which in turn may support later intermedial rebinding at reveal. This suggests a design opportunity: intermedial instruments may benefit from explicit “constraint modes” that intentionally reweight attention and agency, allowing users to move between exploratory play and audiovisually accountable refinement rather than treating constraint purely as an experimental manipulation.

5.2 Interference as cue conflict and productive discontinuity

The results support an interference account centred on cue structure:

- Temporal cue conflict (persistence versus articulation)
- Density cue conflict (visual occupation versus auditory busyness)
- Auditability limits (masking and indistinguishable events)
- Salience dominance (one modality monopolises attention)
- Rebinding (post-reveal reinterpretation that revalues mismatch)

Mismatch was not uniformly negative. Post-reveal replay often revalued apparent mismatch as coherent structure, indicating that interference valence depends on whether divergence produces interpretive dead ends (loss of legibility) or rewarded rebinding (expanded causal model). For practice-based research, this supports treating the reveal as a deliberate compositional and evaluative mechanism.

5.3 Design implications for state-space projection instruments

Because interference manifested as cue reliability problems, design improvements can be framed as interventions on cue structure rather than superficial polish:

- Align temporal envelopes across modalities (for example, link light decay to note release).
- Increase auditory event discriminability (articulation, voicing, timbral differentiation) to reduce masking.

- Strengthen mapping legibility (colour fidelity, persistent reference cues, and clearly communicated constraints).
- Make constraint regimes transparent (loop semantics, reset affordances, and parameter scaling that matches perceptual intuition).

These priorities generalise beyond this case study because they target how performers construct and test causal models while composing.

5.4 Limits and scope

With nine participants, a single synthesis configuration, and a constrained parameter set, the study supports analytic rather than statistical generalisation. We do not claim population-level effects; the contribution is foundational. The article shows that media equality and intermedial interference can be operationalised within realistic artistic development timelines using modality manipulation, replay-based reflection, and traceable artefact capture, and the resulting method is ready to scale. Future work can extend it to other coupled instruments, longer-form composition, and independent-observer evaluation, with sample sizes that license stronger inferential claims.

6 Conclusion

This article operationalised media equality and intermedial interference in a practice-based evaluation of the Cellular Au-Tonnetz, a coupled generative instrument in which sound and LED activity are parallel projections of a shared evolving state. Audiovisual access was most often ranked best overall and most intermedial, while audio-only access most often produced the biggest mismatch.

Across conditions, interference was experienced as both constructive and destructive. Destructive effects were commonly described as cue conflicts (temporal persistence, density correspondence, and perceptual masking), while constructive effects emerged when shared causal origin supported reinforcement and when post-reveal replay enabled rebinding and increased perceived coherence. A central implication is that causal coupling alone does not guarantee perceptual parity: co-equal media relations depend on cue reliability and interaction design that supports evolving causal understanding.

More broadly, the study shows how media equality and intermedial interference can be evaluated through structured practice rather than through descriptive accounts alone. By linking structured reflection to reproducible artefacts (saved presets and recorded outputs), the method supports issue-driven analysis of how media relationships are composed and perceived in coupled generative systems. As foundational practice-based work, it establishes a method that larger-scale follow-ups can refine and extend across other coupled instruments and longer-form composition.

Supplementary Materials

All 32 videos, parameter dataset, and survey results, can be found here:

<https://github.com/td0034/OrganisedSound/>

Acknowledgements

I thank the participants who gave their time to this study, Simon Jones and Sabine Hauert for encouragement to conduct the evaluation, and Rosa Beesley for ongoing support.

The author used ChatGPT (OpenAI) to assist with drafting, language editing, proofing, survey web development, and Python scripting for results analysis. All outputs were reviewed and edited by the author, who takes full responsibility for the final manuscript and materials.

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